



MINING DEALER BEST PRACTICE

C175 Engine Oil Change

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

There is an opportunity for contamination to enter the clean side of the oil system on 793F, 795F and 797F during a routine engine oil PM. If the oil filter canister covers are not sufficiently clean, the contamination left on the covers can enter the clean side of the filters causing reduced engine life. We have witnessed these covers not being cleaned and know that debris will enter into the clean side of the oil system. This contamination will damage the rod and main bearings, but by how much is dependent on the size and amount of debris in the cap.

In addition, when removing the oil pump suction screen, contamination can enter into the system while the screen is being inspected and cleaned.

This Best Practice aims at reducing the chance for contamination as well as reducing the amount of time required to complete the PM through the use of swing components. Utilizing the "swing" cover approach will save around 10-15 minutes per PM, but the big advantage is in quality of the PM.

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2.0 Best Practice Description

For the benefits to be realized in PM efficiency, CC compliance, and achieving engine life, the dealer or customer will need to have at least one extra set of oil filter canister covers (with plugs) and an oil pump suction screen. Good Contamination Control practices recommend these components be placed in sealed plastic bags and delivered to the bay as part of a Standard PM Kit (BP number 1007-2.0-1105). When the new engine oil filter elements are installed a technician should place a new O-ring seal onto the clean replacement covers, and then install onto the oil filter canister. After removal of the oil pump suction screen, the clean replacement screen can be installed along with the cover.

Oil Filter Gp 339-3301

Part Number	Description	Quantity
339-3673	Cover	2
228-7105	Seal	1
8C4424	Plug	1

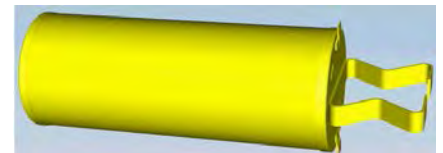


Oil Filter Gp 445-4930

Part Number	Description	Quantity
395-1774	Cover	2
395-1775	Seal Kit	2
228-7100	Seal	2
228-7105	Seal	2
260-9210	Plug	2
8C4424	Plug	2



Part Number	Description	Quantity
364-8521	Screen	1



3.0 Implementation Steps

- Procure a spare set of oil filter canister covers, plugs and engine oil suction screen
- Determine method for cleaning and bagging of components
- Implement cleaned covers and suction screen into the engine oil PM kit

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4.0 Benefits

- Reduction of contamination entering the engine lube system
- Reduced time spent cleaning components
- Suction screen will be cleaned and material will be inspected by a qualified technician for more consistent interpretation
- Mine sites vary from \$5K to \$25K per truck per hour. If you use a conservative \$10k per hour that is \$166 per min or \$1666 every 10 minutes. So you can see that in time savings alone you will pay for the swing components on the first PM on the first truck. Currently there is no way to quantify how much you are shortening the life of the engine if you choose to not use swing covers and screen.

5.0 Resources Required

- The cost of the filter covers is \$450 x 2 and the suction screen is \$130 or \$1,030USD total.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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